Assignment 1 Solution

```
install.packages("marginaleffects")
```

```
Installing package into '/usr/local/lib/R/site-library'
(as 'lib' is unspecified)
```

```
library(tidyverse)
```

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✔ dplyr     1.1.4     ✔ readr     2.1.5
✔ forcats   1.0.0     ✔ stringr   1.5.1
✔ ggplot2   3.5.0     ✔ tibble    3.2.1
✔ lubridate 1.9.3     ✔ tidyr     1.3.1
✔ purrr     1.0.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to b
```

```
library(broom)
library(marginaleffects)
```

```
my_url <- "http://ritsokiguess.site/datafiles/roller.csv"
roller <- read_csv(my_url)
```

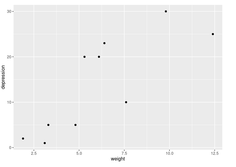
```
Rows: 10 Columns: 2
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
dbl (2): weight, depression

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
roller
```

```
# A tibble: 10 × 2
   weight depression
    <dbl>      <dbl>
 1    1.9          2
 2    3.1          1
 3    3.3          5
 4    4.8          5
 5    5.3         20
 6    6.1         20
 7    6.4         23
 8    7.6         10
 9    9.8         30
10   12.4         25
```

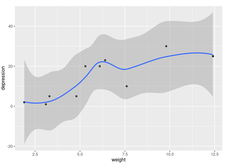
Two quantitative variables, so we need a scatterplot. The weight of the roller is explanatory (input), and the depression caused by the roller is the response (output), hence:

```
ggplot(roller, aes(x = weight,  y = depression)) + geom_point()
```




```
ggplot(roller, aes(x = weight, y = depression)) + geom_point() + geom_smooth()
```

```
`geom_smooth()` using method = 'loess' and formula = 'y ~ x'
```



The trend of the points is more or less linear. Likewise, if we added a smooth trend, it wiggles a bit but there is no obvious curved shape.

```
roller.1 <- lm(depression ~ weight, data = roller)
summary(roller.1)
```

```
Call:
lm(formula = depression ~ weight, data = roller)

Residuals:
   Min     1Q Median     3Q    Max
-8.180 -5.500 -1.346  5.920  8.020

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  -2.8071     4.7543  -0.439  0.67227
weight        2.6667     0.7002   3.808  0.00518 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 6.735 on 8 degrees of freedom
Multiple R-squared:  0.6445,	Adjusted R-squared:  0.6001
F-statistic:  14.5 on 1 and 8 DF,  p-value: 0.005175
```

There is a moderately high R-squared (suggesting a real relationship with weight, but one with a certain amount of scatter). This is supported by the small P-value (0.005) for weight , and the slope of 2.67 is positive, indicating an upward trend.

```
new <- data.frid(model = roller.1, weight = seq(2.5, 12.5, 2.5))
new
```

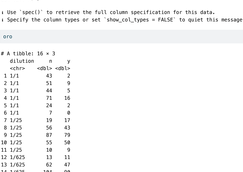
```
  depression weight
1       14.1    2.5
2       14.1    5.0
3       14.1    7.5
4       14.1   10.0
5       14.1   12.5
```

```
cbind(predictions(roller.1, newdata = new)) %>%
  select(weight, estimate, starts_with("conf"))
```

```
  weight  estimate  conf.low conf.high
1    2.5 4.579717 -1.857207  11.01064
2    5.0 11.246582  6.821281  15.67196
3    7.5 17.913447 13.300503  22.52639
4   10.0 24.580311 17.759772  31.40085
5   12.5 31.247176 21.484722  41.00963
```

The intervals seem to be wide due to the small number of observations.

```
plot_predictions(roller.1, condition = "weight")
```



The predictions form a line, and the envelope around the line is the confidence interval for a prediction at each weight. The confidence intervals are narrower in the middle, because there are more "nearby" points to base a prediction on.

In question 4, the confidence interval went from 17.75 to 31.4. On the graph, the interval should correspond to the bottom and top of the envelope at that value of weight . For a weight of 10, the bottom of the envelope is just over halfway from 15 to 20 (that is, just over 17.5), and the top of the envelope is less than halfway from 30 to 35 (so, less than 32.5). These are consistent with the values from question 4.

```
my_url <- "http://ritsokiguess.site/datafiles/aod_orob1.csv"
oro <- read_csv(my_url)
```

```
Rows: 16 Columns: 3
── Column specification ────────────────────────────────────────────────────────
Delimiter: ","
chr (1): dilution
dbl (2): n, y

ℹ Use `spec()` to retrieve the full column specification for this data.
ℹ Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
oro
```

```
# A tibble: 16 × 3
   dilution     n     y
   <chr>    <dbl> <dbl>
 1 1/1         43     9
 2 1/1         51     9
 3 1/1         44     5
 4 1/1         71    16
 5 1/1         24     2
 6 1/1          7     0
 7 1/25        19    17
 8 1/25        56    43
 9 1/25        87    79
10 1/25        55    50
11 1/25        10     9
12 1/625       13    11
13 1/625       62    47
14 1/625      104    90
15 1/625       51    46
16 1/625       11     9
```

The observations in this case are seeds (each one either germinates or not, which is the actual response), and there is more than one seed per row, specifically the number in the column n . (If there had only been one seed per row, there would have been a column with a name like germinated and values "yes" and "no", or "true" and "false").

We have "grouped" observations (more than one per row), so we need a two-column response, the number of successfully germinated seeds in the first column (which is y), and the number of seeds that failed to germinate in the second (which you will have to include a calculation for):

```
oro.1 <- glm(cbind(y, n - y) ~ dilution, family = "binomial", data = oro)
summary(oro.1)
```

```
Call:
glm(formula = cbind(y, n - y) ~ dilution, family = "binomial",
    data = oro)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)   -1.8015     0.1851  -9.732  <2e-16 ***
dilution1/25   3.7225     0.2717  13.703  <2e-16 ***
dilution1/625  3.4771     0.2559  13.585  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 387.689  on 15  degrees of freedom
Residual deviance:  22.945  on 13  degrees of freedom
AIC: 78.273

Number of Fisher Scoring iterations: 4
```

Having a categorical explanatory variable means that the two (very significant) tests shown in the summary output are of the level shown against the baseline 1/1 dilution, and not what we want. Hence, we need a drop1 , as in regular regression. Since this is a generalized linear model, so the test needs to be Chisq rather than F.

```
drop1(oro.1, test = "Chisq")
```

```
Single term deletions

Model:
cbind(y, n - y) ~ dilution
       Df Deviance  AIC  LRT Pr(>Chi)
<none>     22.94 78.27
dilution  2  387.69 439.42 364.74 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

With a tiny P-value of 6.3 × 10−80 (!), there is definitely an effect of dilution overall.

Another option is anova (i.e. for comparing two models). Compare the model with dilution (the one we fitted) against the model without. The answer is to have a 1 on the right side of the squiggle:

```
oro.0 <- glm(cbind(y, n - y) ~ 1, family = "binomial", data = oro)
```

```
anova(oro.0, oro.1, test = "Chisq")
```

```
Analysis of Deviance Table

Model 1: cbind(y, n - y) ~ 1
Model 2: cbind(y, n - y) ~ dilution
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1        15     387.69
2        13      22.94  2  364.74 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Comparing the output here vs. the drop1 , it is evident that the test and P-value are exactly the same.

```
oro %>% count(dilution) -> new
new
```

```
# A tibble: 3 × 2
  dilution     n
  <chr>    <int>
1 1/1          6
2 1/25         5
3 1/625        5
```

```
cbind(predictions(oro.1, newdata = new))
```

```
  rowid  estimate      p.value  s.value  conf.low conf.high dilution n
1     1 0.1146667 2.2001D-22 71.94479 0.1030001 0.1917459      1/1 6
2     2 0.8722467 4.3933D-22 70.94545 0.8221924 0.9097569     1/25 5
3     3 0.8423237 2.5436E1-21 68.41360 0.7907064 0.8838929    1/625 5
```

```
cbind(predictions(oro.1, newdata = new)) %>%
  select(dilution, estimate, conf.low, conf.high)
```

```
  dilution  estimate  conf.low conf.high
1      1/1 0.1146667 0.1030001 0.1917459
2     1/25 0.8722467 0.8221924 0.9097569
3    1/625 0.8423237 0.7907064 0.8838929
```

In question 10, we found there was a (very) significant effect of dilution . The predictions here suggest that this is because the 1/1 dilution is very different from the other two (less germination), which are very similar in terms of germination to each other.

There are two reasons why the 1/25 and 1/625 dilutions are similar in terms of germination: first, their Estimates are very close, but second and more insightfully, their confidence intervals overlap, so from that point of view, they are not significantly different in terms of germination.